3.1 The Chain Rule

You missed this lesson. Copy this into your notes. That is all you need to do.

Question of the day

How fast does a function of a function change?

Targets

- I can name the inside and the outside of a composite function.
- I can differentiate a composite function with the chain rule.
- I can find a slope at a point from a table.

Concepts

- **Composite function** (y = f(g(x))): The output of the inside becomes the input of the outside. The outside never sees x directly.
- **The chain rule** (d/dx f(g(x)) = f?(g(x)) · g?(x)): Differentiate the outside and leave the inside alone. Then multiply by the derivative of the inside.
- **Layers** (dy/dx = dy/du · du/dx): Each layer adds one factor. Peel from the outside in.
- **Reading a table** (h?(a) = f?(g(a)) · g?(a)): Find g(a) first. Read f? at that output, not at a.

What the class predicted, and what is true

- Expand: $y = (3x)^2 = 9x^2$, so the slope is $18x$. The outside rule alone gives $2(3x) = 6x$. What factor is missing? **3, the slope of the inside.** 6x times 3 is 18x. The inside 3x changes at rate 3, and that rate multiplies in.
- The value y changes 2 units for every 1 unit of u. The value u changes 5 units for every 1 unit of x. How many units does y change per unit of x? **10.** The rates multiply: 2 times 5 is 10.

Huddle notes (answers filled in)

Definitions

- **Composite function:** One function fed into another. The **inside** runs first.
- **Chain rule:** Differentiate the **outside**, leave the inside, then multiply by the **inside?**.
- **Reading a table:** To read h?(a), find **g(a)** first.

Worked example

Differentiate $y = \sin(x^2 - 3)$.

- Inside: $u = x^2 - 3$, so $du/dx =$ **2x**
- Outside: sin u, so $dy/du =$ **cos u**
- Multiply: $dy/dx =$ **$2x \cdot \cos(x^2 - 3)$**

Label the assembly line

1. The first machine is the **inside** function.
2. The last machine is the **outside** function.
3. The middle value u is the **output** of the inside.

One worked sample of each skill

Differentiate a composite

Level 1. Differentiate. Leave the inside alone.
Answer: $3\,(x+4)^2$ Outside: bring down 3, lower the power. Inside $x+4$ stays. So $3(x+4)^2$.

Level 2. Differentiate. Multiply by the slope of the inside.
Answer: $10\,(5x+6)$ Outside: $2(5x+6)$. Inside slope: 5. Product: $10(5x+6)$.

Level 3. Differentiate.
Answer: $(-2)\sin(2x+5)$ Outside: $-\sin(2x+5)$. Inside slope: 2. Product: $-2\sin(2x+5)$.

Level 4. Differentiate. Rewrite the root as a power if that helps.
Answer: $\frac{x}{\sqrt{x^2+2}}$ $\dfrac{1}{2\sqrt{x^2+2}}\cdot 2x = \dfrac{x}{\sqrt{x^2+2}}$.

Name the inside

Level 1. In $y=(x+4)^3$, the inside function is...
(A) x + 4 (B) x^3 (C) y (D) 4
Answer: A x goes into $x+4$ first; then the result is raised to the power 3.

Level 2. In $y=\sin(2x)$, which function is the outside?
(A) 2x **(B) sin** (C) 2 (D) x
Answer: B x is multiplied by 2 first. Sine is applied last, so sine is the outside.

Level 3. In $y=e^{\sin x}$, the inside function is...
(A) e? (B) x (C) e^(sin x) **(D) sin x**
Answer: D x goes into sine first. The exponential is applied to that output.

Level 4. How many factors does the derivative of $y=\sin\left(\sqrt{x^2+6}\right)$ have?
(A) Two (B) One **(C) Three** (D) Four
Answer: C Three layers: sine, the square root, and x^2+6. Each gives one factor.

Read a table

Level 1. A table gives $g(2)=3$, $g'(2)=4$ and $f'(3)=5$. Let $h(x)=f(g(x))$. Find $h'(2)$.
Answer: 20 $h'(2)=f'(g(2))\cdot g'(2)=f'(3)\cdot g'(2)=5\cdot 4=20$.

Level 2. A table gives $f'(1)=3$, $g(1)=6$, $g'(1)=-5$ and $f'(6)=2$. Let $h(x)=f(g(x))$. Find $h'(1)$.
Answer: -10 g(1) = 6, so $h'(1)=f'(6)\cdot g'(1)=2\cdot(-5)=-10$. The value $f'(1)$ is a distractor.

Level 3. A table gives $f(4)=1$, $f'(4)=4$, $g'(4)=6$ and $g'(1)=2$. Let $h(x)=g(f(x))$. Find $h'(4)$.
Answer: 8 f(4) = 1, so $h'(4)=g'(1)\cdot f'(4)=2\cdot(4)=8$.

Level 4. A table gives $g(4)=6$, $g'(4)=2$, $f(6)=4$ and $f'(6)=5$. Let $h(x)=f(g(x))$. The tangent line to h at $x=4$ is used to estimate $h(5)$. Find the estimate.
Answer: 14 $h(4)=f(6)=4$ and $h'(4)=5\cdot(2)=10$. One step right gives $4+(10)=14$.

Slope at a point

Level 1. Let $y = (x+2)^2$. Find the slope of the curve at $x = 2$.
Answer: 8 $y' = 2(x+2)$, so at $x = 2$ the slope is $2 \cdot 4 = 8$.

Level 2. Let $y = (4x+3)^2$. Find the slope of the curve at $x = 1$.
Answer: 56 $y' = 8(4x+3)$, so at $x = 1$ the slope is $8 \cdot 7 = 56$.

Level 3. Let $y = \ln(x^2+8)$. Find the slope of the curve at $x = 1$.
Answer: 0.2222 $y' = \dfrac{2x}{x^2+8}$, so at $x = 1$ the slope is $\dfrac{2}{9}$.

Level 4. Let $y = \sqrt{x^2+33}$. Find the slope of the curve at $x = 4$.
Answer: 0.5714 $y' = \dfrac{x}{\sqrt{x^2+33}}$, so at $x = 4$ the slope is $\dfrac{4}{\sqrt{49}} = 0.5714$.